ISBN: 978-3-95450-263-9 ISSN: 2673-5520 doi: 10.18429/JACoW-MEDSI2025-TUP18

DESIGN OF A DEDICATED MULTILAYER MONOCHROMATOR FOR ELECTRON BEAM SIZE MEASUREMENTS USING THE HETERODYNE NEAR FIELD SPECKLES (HNFS) TECHNIQUE AT THE ALBA SYNCHROTRON

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Abstract

Within the framework of the ALBA Diagnostics Group's participation in a Future Circular Collider (FCC) collaboration, a dedicated setup for electron beam size measurement based on Heterodyne Near Field Speckles (HNFS) has been developed and installed at ALBA Front End 21 using radiation from a dipole bending magnet. The setup incorporates a high-energy (20–30 keV), high-bandwidth (~1.3%) monochromator, entirely designed in-house, along with the colloid sample environment and the detector system with their corresponding supports. The monochromator features a 300 mm Si substrate with W/B_4C multilayer coating and operates in a vertical Laue reflection geometry. To reduce complexity and for this HNFS-specific application, ultra-high vacuum (UHV) conditions and submicron precision mechanics are not required. The mirror assembly is housed within a standard DN400 CF chamber, mechanically coupled to the chamber itself. This chamber is mounted on a granite-based "skin concept" table, providing two degrees of freedom (vertical translation and tilt) for energy tuning and beam path insertion/retraction. The complete design of the set-up is presented in this paper.

INTRODUCTION

ALBA Beam Physics group in collaboration with CERN and the University of Milano is developing a technique to measure storage ring electron beam size, the so-called Heterodyne Near Field Speckles (HNFS). After first successful measurements using HNFS at ALBA BL11 NCD beamline [1], a dedicated set-up to conduct these measurements has been designed and installed at Fron End 21 (FE21) [2]. Unlike the previous set-up at BL11, FE21 is illuminated by a dipole bending magnet and currently is used by ALBA diagnostics group to conduct beam size measurements using the pinhole technique. This allows direct comparison between both techniques.

The whole set-up has been fully developed at ALBA and consists of a monochromator and a sample and detector table. Both components can be operated to be fully retracted from the x-ray beam path, allowing the alternative use of pinhole diagnostics. A schematic diagram of HNFS set-up integration in FE21 is depicted in Fig. 1.

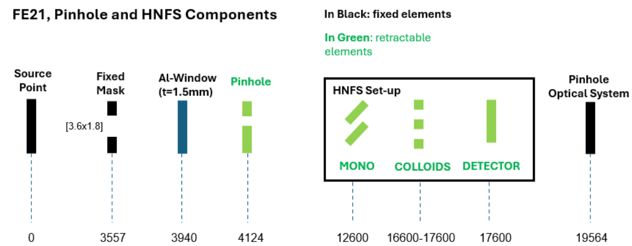

Figure 1: HNFS set-up integration in ALBA FE21 with relative distances from source point.

GENERAL DESCRIPTION

Due to HNFS experimental nature, consisting of x-ray diffracting upon colloid suspension samples, the design of monochromator and sample and detector support is oriented to build simple and cost-effective devices rather than to achieve high end performances in terms of motion accuracy and stability (see Fig. 2).

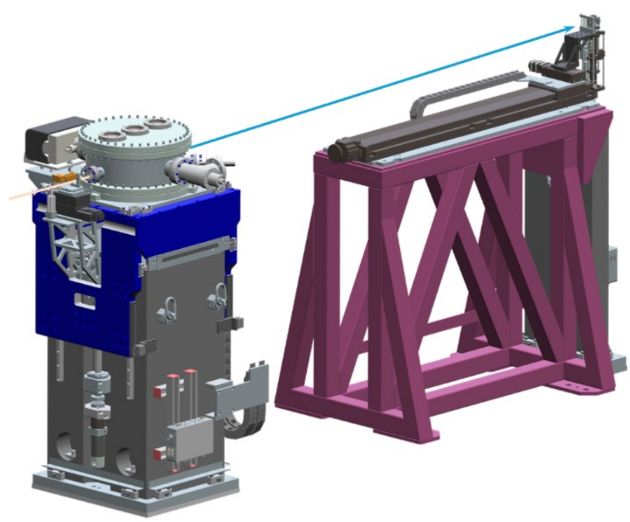

Figure 2: External view of HNFS Monochromator and Detector and Sample table.

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TECHNICAL SPECIFICATIONS

Considering FE21 photon source and HNFS technique characteristics, the monochromator must comply with the following specifications:

- Energy range: 20-30 keV
- Bandwidth, ΔE/E ~ [1-2%]
- Maximize photon flux. Selection of multilayer Ru/B4C mirror.
- LAUE monochromator geometry.
- Crystal oriented to deviate the beam in the vertical plane.
- UHV conditions are not required.
- Possibility to fully retract the instrument out of the x-ray beam.

Support and Motion Stages

The monochromator support and motion mechanics are based in the well-known skin concept architecture [3]. This concept allows to control with a compact design two DOF: pitch angle, to tune Monochromator energy, and vertical direction, to fine adjustment of vertical position and to allow full retraction/insertion of the device from the x-ray beam. These movements are actuated by two low-backlash stepper motors that result in a pseudo-motor and include absolute encoder feedback. Moreover, the design perfectly adapts to the required tilt range and resolution, see Table 1, while keeping a compact and rigid configuration. This is accomplished thanks to approaching the optic components as much as possible to the granite base, see Fig. 3, the use of flexure joints between rotating parts and the selection of high precision preloaded mechanical components such as: linear guides and ball screws.

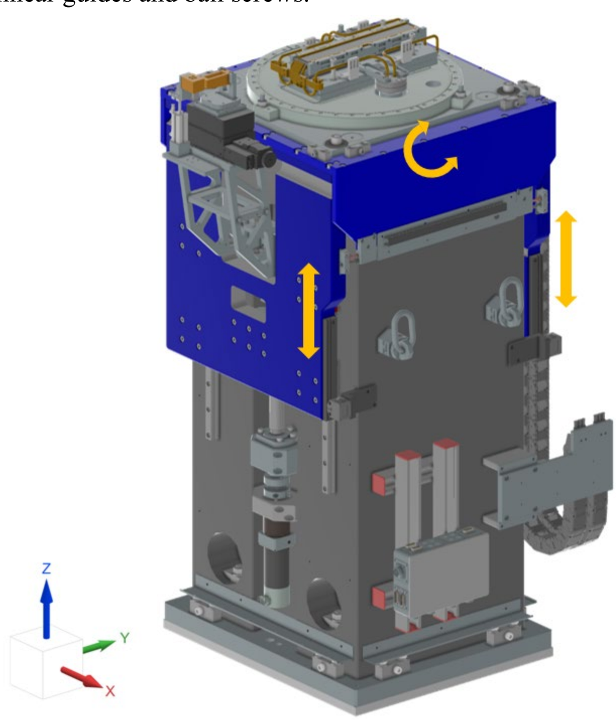

Figure 3: Monochromator skin concept table motion mechanics and support.

Table 1: Monochromator Motions Compliance Table.

		Required	Accomplished
Z	Range	+5/-160 mm	+5/-160 mm
	Resolution	1 µm/fs	0.25 µm/fs
Pitch	Range	0,64± 0,3 deg	± 0,3mrad
	Resolution	1 µrad/fs	0.44 µrad/fs

Considering the monochromator mirror length and the nominal X-ray beam height of 300 mm and 1400 mm, respectively, the support and motion mechanics result in a highly compact construction with overall dimensions of 605× 613×1280 mm, in the Z full retracted position.

Taking advantage of the design's compactness and stability, and with the aim of simplifying the design and reducing project costs, the vacuum chamber and monochromator optics are not decoupled. Instead, they are supported and moved together by the skin concept table.

Multilayer Mirror Environment

The mirror substrate holder is mounted on the vacuum chamber base plate and consists of a kinematic mount preloaded with springs, which provides fine adjustment and repeatability in pitch and roll (see Fig. 4). The mirror itself rests on three ball-point contacts and is secured in position by elastic clamps. The holder also supports symmetric copper-pipe water-cooling circuits, which are clamped against the mirror substrate using six threaded rods with elastic springs, ensuring a distributed and controlled contact force. Each copper cooling circuit is brazed to a DN40 CF flange, assembled to a vacuum-welded bellow, and connected to the vacuum chamber base plate. In addition, a protective chin guard is fixed to the mirror holder and connected to the cooling circuit by two copper braids. Four Pt100 temperature sensors have been incorporated near the irradiated surface of the mirror. Overall, the design of the mirror holder and cooling system is inspired by the MINERVA beamline monochromator, another ALBA in-house development [4].

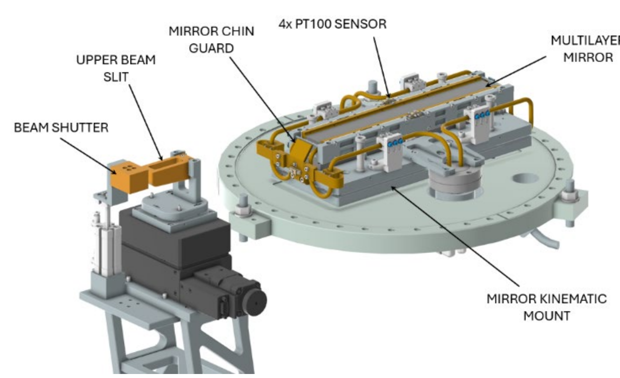

Figure 4: Mulilayer mirror environment, upper slit and beam shutter.

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Outside the vacuum chamber, on the incoming side of the monochromator, the upper beam slit and the beam shutter are mounted on a support attached to the granite base. The upper beam slit consists of a wedged copper element driven by a commercial elevation stage, whose position depends on the operating energy of the monochromator. Its function is to shape the upper portion of the incoming X-ray beam, preventing part of it from spilling beyond the active mirror surface and passing through the monochromator.

Static and Modal Analysis Simulations

Different calculations by means of FEA method have been done in order to achieve the standards of safety and to comply with the technical specifications of the project. In the field of structural analysis, the stresses in the flexures have been checked when extreme angular position is achieved. The resulting value of 115 MPa is well under control with standard structural steel.

Since the optical elements are mechanically coupled to the rest of the system, several finite element analyses (FEA) were performed to study the free vibration behaviour of the monochromator. After several iterations, the first resonance mode (RM) was identified at 81 Hz, corresponding to a pitch mode, as shown in Fig. 5. The RM frequencies are summarized in Table 2, supporting the design choice of not decoupling the optical mechanics from the vacuum component support.

Table 2: Monochromator Motions Compliance Table.

Mode	Frequency (Hz)
1	81.4
2	101.2
3	117.3
4	150.9
5	153.2
6	172.0

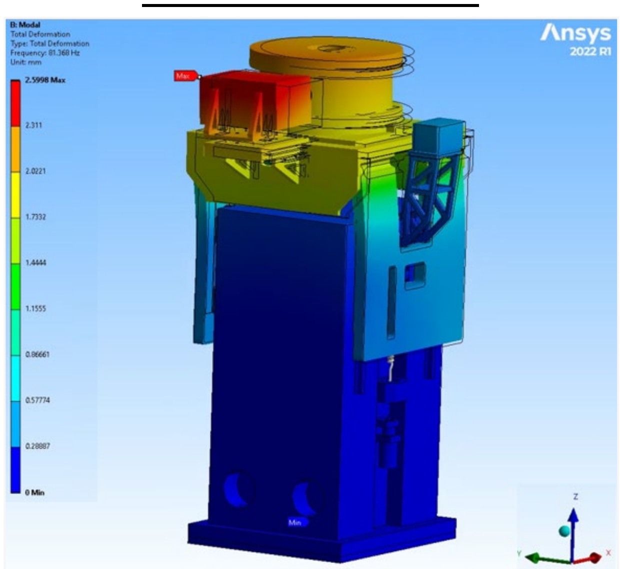

Figure 5: 1ST pitch RM.

Thermal Simulations

Considering the worst-case scenario in terms of thermal load density, taking place when the monochromator is operated at 20 keV, which produces a total absorbed power of 17 W, a set of thermal simulations have been done to assess the mirror temperature distribution and the thermal induced deformations, see Fig. 6, obtaining values within our acceptance criteria.

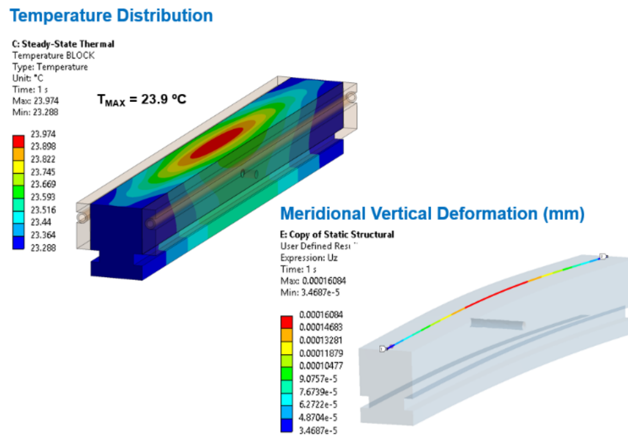

Figure 6: Temperature distribution and meridional vertical deformation.

Figure 7: Monochromator and sample table during installation at FE21.

CONCLUSION

A dedicated setup for storage ring beam size measurements using the HNFS technique has been designed in-house at ALBA. Installed during the last summer shutdown, as illustrated in Fig. 7, the system is scheduled to begin commissioning in the coming weeks.

This paper has focused primarily on the design of the monochromator, which builds upon previous ALBA developments [1, 2] to deliver a simple and cost-effective device tailored to the requirements of the dedicated HNFS experimental setup.

ACKNOWLEDGMENTS

The authors wish to acknowledge all those most involved in the project: José Ferrer, Karim Maimouni, Diego Rubia, Jon Ladrera, Dominique Henis, Josep Nicolas and Albert Van Eeckhout.

ISBN: 978-3-95450-263-9 ISSN: 2673-5520 doi: 10.18429/JACoW-MEDSI2025-TUP18

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